Software Tools and Technologies Laboratory
Latex Lab Activity - 3: Beamer

Anand M. Baswade Arindam Chakraborty

Faculty of Computer Science and Engineering
Indian Institute of Technology, Bhilai
M.Tech Student of Computer Science and Engineering
Indian Institute of Technology, Bhilai

15th September, 2025

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2. Why Beamer?
3. Features
4. Highlighting Texts
5. Mathematical Concepts
6. Linear Search Algorithm
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Introduction to LaTeX Beamer

- LaTeX Beamer is a powerful class for creating presentations.
- It allows:
  - Professional-quality slides
  - Mathematical equations
  - Figures, tables, and references
  - Animations and overlays
- Advantages:
  - Consistent formatting
  - Easy handling of citations and references
  - Cross-platform PDF output

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- Free and open-source — no licensing issues

Comparison with Other Tools

Feature	Beamer vs PowerPoint
Equations	Excellent vs Limited
Citations	Automatic vs Manual
Portability	PDF (stable) vs File-dependent
Cost	Free vs Proprietary
Customization	High (code-based) vs Moderate (GUI-based)

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- Integration with BibTeX for automatic citations

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Blocks and Highlights Example

Remark

This is a normal block. It wraps the text in a box with the same style as the presentation.

Important Theorem

This is an alert block. It contrasts with the regular presentation style.

Examples

This is an example block. Perfect for examples.

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Definition

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A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself.

Theorem & Lemma

Theorem

Fundamental Theorem of Arithmetic: *Every integer greater than 1 is either a prime or can be written as a unique product of prime numbers.*

Lemma

If a prime p divides a product ab, then p divides a or p divides b.

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Linear Search Algorithm

Algorithm 1 Linear Search

```
1: procedure LinearSearch(A, key)
2:     for i = 0 to n − 1 do
3:         if A[i] = key then
4:             return i
5:         end if
6:     end for
7:     return -1
8: end procedure
```

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Corollary & Proof

Corollary

The factorization of any integer into primes is unique up to the order of the factors.

Proof. Suppose n has two different prime factorizations. By the lemma, every prime in one factorization divides one of the primes in the other factorization. This leads to a contradiction, proving uniqueness.

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Including an Image in a Block

Example Image

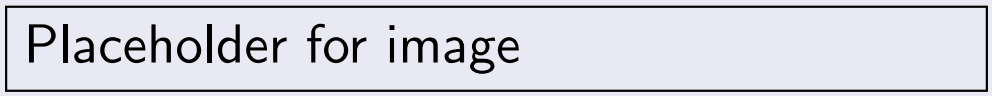


Figure 1: This is a sample image included in a Beamer slide.

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Key Equations in Signal Processing

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft}\, dt$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau)\, d\tau$$

$$H(f) = \frac{Y(f)}{X(f)} = \int_{-\infty}^{\infty} h(t) e^{-j2\pi ft}\, dt$$

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- Overlays and incremental reveals help guide the audience's focus.
- Highly customizable themes, colors, and backgrounds make slides visually appealing.

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Any Questions?